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EXAMEN CODE:

**Specific instructions for the Chemistry Contest**

Highly esteemed student:

- This exam was created by a Scientific Committee composed of former olympic students who work (mostly) at the University of Havana.
- **Complete your personal information.** Please verify with your ID card.
- The direction of the Provincial Education is responsible for ensuring that the exam is printed in a sufficiently high quality, allowing students to read the instructions, questions, figures, data, graphs, and tables. **No questions will be canceled due to poor printing.**
- Write all your information clearly in the space provided (first name, last name, school, municipality, province). **Do not write anything in the CODE field,** it is for the exclusive use of the coding manager.
- **Check that your exam is complete.** In the case of the exam on the first day, 10th grade should answer problems 7, 8, 9 and 10; 11th grade should answer problems 7, 8, 9 and 11; 12th grade should answer problems 7, 11 and 12.
- All necessary constants, conversions, equations, and periodic table are included on the first page of the exam. This page will be removed during the coding process.
- At the beginning of each question, the grading is indicated, its value (out of a total of 100 points), as well as the value of each item (in marks).
- You can use a scientific calculator and a ruler.
- Please comply to the rules of approximate calculation. Attention! You have limited time and the exam has a large number of items. You do not have a space limit to answer, BUT please answer in an orderly manner, with appropriately sized and legible font, to facilitate the grading process.
- You must present all auxiliary calculations, justifying your reasoning and answers properly whenever necessary. **Unintelligible answers will not be graded.**
- Number the additional answer sheets at the bottom of each page and indicate the total number of pages, for example, “1 of 7,” “2 of 7,” and so on.
- If the exam instructor/proctor interferes with any of the above guidelines due to ignorance, exercise your right to protest and immediately request the intervention of the competent authority of the Ministry of Education, which is the institution responsible for administering the exam.
- The President of the Scientific Committee can be reached at +53-5045-4299.

You are a source of pride for Cuban education: Best of luck!

EXAM CODE (only for the use of the code manager)

Avogadro's Constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal Gas Law	$p \cdot V = n \cdot R \cdot T$
Gas constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$ $= 0,08205 \text{ atm} \cdot \text{L} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta U + W$
Zero in Celsius scale	273,15 K	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{c_{\text{red}}}{c_{\text{ox}}} \right)$
Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbance	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Vapor pression lowering	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Sphere volume	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Sphere area	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Circle area	$A_{\text{sp}} = \pi \cdot r^2$	Speed of light	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Length of a circumference	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Gravity force	$F_g = m \cdot g$	Gravity on Earth	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$ $1 \text{ bar} = 100 \text{ kPa}$ $1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$ $1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$ $1 \text{ dm}^3 = 1 \text{ L}$ $1 \text{ mL} = 1 \text{ cm}^3$ $1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																	2 He 4.003
2	3 Li 6.941	4 Be 9.012											5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305											13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	* 72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	** 104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]

*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]

EXAM CODE (only for the use of the code manager)

Problem 1. Hydrides (10% of total; 10th, 11th and 12th grades)

1.1	1.2	1.3	1.4	1.5	1.6	Total	Total
11	7	10	2	5	5	40	10%

One day, Lauren and Leticia were preparing a lesson about hydrides for tenth-grade students. Lauren wrote the following statement. Can you fill in the blanks to make it understandable?

1.1. Hydrogenated binary compounds can be classified into four main types. i) _____ hydrides are formed with the least electronegative metals. They have an _____ chemical bond and a large _____ partial charge on the hydrogen atom. They react vigorously with water, releasing _____, which reflects their potent _____ and _____ characteristics. ii) _____ hydrides are formed when hydrogen atoms occupy the empty spaces in the crystal lattices of the transition elements. They have chemical properties like the previous ones, although more attenuated, because in them the partial charge on the hydrogen atom is _____. They are electrical conductors in the solid state because the chemical bond they possess is _____ metallic _____. iii) _____ hydrides are formed with some chemical elements from groups 2, 11, 12, and 13, which have lower electronegativities and empty orbitals. The chemical bond in this case has intermediate characteristics. These hydrides react readily with the non-bonding electron pairs of Lewis _____. iv) Volatile hydrides are formed with non-metals and their acid-base behavior varies. Group 15 hydrides behave as bases because they have _____ in their molecular structure. For example, ammonia in aqueous solution increases the concentration of hydroxide ions in water. In contrast, group 16 and 17 hydrides in aqueous solution behave as acids because they have a large partial _____ on the hydrogen atom, thereby increasing the concentration of _____ ions in water. They react vigorously with alkali metals, releasing _____, which reflects their acidic and oxidizing characteristics. Hydrides of the first three types are _____ solids _____ at room temperature, while most type iv hydrides are _____, with _____ being a notable exception. Hydrogenated binary compounds of carbon, known as _____, can exist in all three aggregation states: as the chain length increases, the melting and boiling points _____ due to an _____ in _____ interactions.

For her part, Leticia found that ammonia has anomalous properties compared to the other hydrides in group 15, as shown in the following table.

Hydride	NH ₃	PH ₃	AsH ₃	SbH ₃	BiH ₃
Teb (°C)	-33	-88	-63	-17	17
α H-X-H (°)	107	93	92	92	91

1.2. Explain in detail why

- The boiling point of ammonia is unusually high.
- The bond angle is close to 90° in the hydrides mentioned above, except for ammonia.

Leticia and Lauren decided to demonstrate a practical application of a volatile hydride. Under a fume hood, they set up the equipment shown in the figure to generate a certain toxic and odorous gas *in situ*, which they then bubbled into 1,0 mol/L aqueous nitrate solutions.

The results obtained were as follows:

$\text{Al}(\text{NO}_3)_3$: No change was observed.

$\text{Fe}(\text{NO}_3)_3$: A black precipitate and a yellowish-white turbidity formed in the solution.

$\text{Co}(\text{NO}_3)_2$: A black precipitate formed and the pink solution became discolored.

$\text{Cu}(\text{NO}_3)_2$: A black precipitate formed and the blue solution became discolored.

$\text{Pb}(\text{NO}_3)_2$: A black precipitate formed.

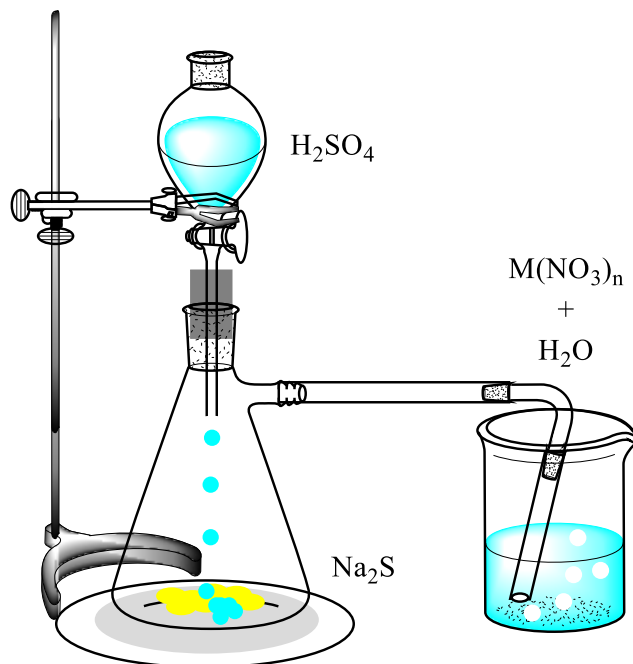
All the final solutions became acidic, although the first was the least acidic of all.

1.3. Formulate balanced chemical equations (ionic or molecular) corresponding to all reactions that occurred in the procedures described. Indicate the states of aggregation.

1.4. Why are Fe^{3+} , Co^{2+} and Cu^{2+} ions colored in aqueous solution?

1.5. Both contestants were struck by the fact that sulfide anions have a greater affinity for the cations Fe^{2+} , Co^{2+} , Cu^{2+} and Pb^{2+} than for the cations Al^{3+} and Fe^{3+} . Why?

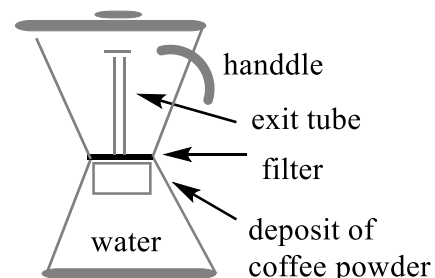
1.6. Identify the laboratory tools shown in the figure.



Problem 2. A can of coffee
(15% of total; 10th, 11th and 12th grades)

2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	Total	Total
2	2	5	3	18	6	2	10	12	60	15%

Most Cubans enjoy drinking coffee. If you visit a home or institution, you'll likely be offered a white porcelain cup containing about 50 mL of the beverage. Depending on the location, coffee can be heated with electricity, natural gas, coal, or firewood. Typically, coffee is prepared in a coffee pot, in an energy-intensive process. First, the water must be brought from room temperature, 25°C, to the boiling point, 100°C, consuming energy amount Q_1 . Then, a small amount of water turns into a gaseous state, causing an increase in pressure inside the lower reservoir and consuming energy amount Q_2 . Liquid water then mixes with coffee powder and extracts the main water-soluble component of coffee, caffeine; the formation of bubbles aids the extraction process. The excess pressure then causes the resulting solution to escape through the outlet tube. Finally, all that remains is to wait for the coffee to cool slightly and enjoy the beverage. A four-cup Cuban coffee pot can contain up to 200 mL of water; however, at the end of a "colada", only about 190 mL of coffee is obtained, which can be attributed primarily to evaporation losses.



To answer all the following questions, consider coffee to be an ideal dilute solution of the solute caffeine in the solvent water. Therefore, the heat capacity and density of coffee are equal to those of pure water, respectively, $c = 4,18 \text{ J}\cdot\text{K}^{-1}\cdot\text{g}^{-1}$ and $\rho = 1,00 \text{ g/mL}$. Also, use the thermodynamic data provided at the end of the question.

2.1. Coffee pots are made of metal, which facilitates both rapid heating and cooling. Why are metals good conductors of heat?

2.2. How much energy Q_1 (kJ) is required to heat all the water inside a full coffee pot?

2.3. How much energy Q_2 (kJ) is required to evaporate 10 mL of water?

2.4. What amount (mole) of water vapor exists inside the lower container when the last portion of the liquid is coming out? Assume that water vapor behaves in an ideal manner.

2.5. is the minimum amount of electricity (moles of electrons), natural gas (L at standard ambient conditions), coal (kg), and wood (kg) required to brew coffee in a coffee pot 1000 times?

In the village of Jatibonico, mornings are cold, typically around 15°C . Mario enjoys drinking coffee in the morning, but in his village, the beverage must be insulated in a thermos to keep it hot for a long time. The organizers gave all participants of the 56th ICHO a 1,0 L thermos, which Mario uses to store coffee. The heat flux Q/t can be calculated using the following equation: where k is the thermal conductivity coefficient of the material, A is the cross-sectional area through which heat energy transfer occurs, l is the thickness of the material, and ΔT is the temperature difference.

$$\frac{Q}{t} = \frac{k \cdot A \cdot \Delta T}{l}$$

2.6. Estimate the minimum time t (in min) required for all the freshly brewed coffee inside Mario's thermos to cool down to 15°C . Consider the cylindrical thermos (diameter 10,0 cm; height 14,0 cm) made of a material with $k = 0,15 \text{ W}\cdot\text{m}^{-1}\cdot^\circ\text{C}^{-1}$ and a thickness of 0,50 cm.

2.7. In practice, the cooling time is longer than you calculated. Why?

Emmanuel and *Riky* live in the city, but they enjoy hiking, biking, and overnight camping. Self-heating containers are more useful than thermoses. By pressing a button, they can break a film, separating a suitable solid chemical from the water in a small reservoir inside the container. An exothermic process then occurs, heating the entire container. The solid can be composed of a pure substance or a mixture, for example,

CaO , CaCl_2 , $\text{Mg} + \text{Fe}$, $\text{AlCl}_3 + \text{CaO}$, $\text{P}_2\text{O}_5 + \text{CaO}$, $\text{CuSO}_4 + \text{Zn}$.

2.8. Formulate the equations corresponding to the reactions that occur (if any) in each case. In the third formulation, the iron acts as a redox catalyst; it is not consumed in the process.

2.9. One of the products formed, calcium hydroxide, has a high enthalpy of formation, which makes the process exothermic. Estimate the value of this magnitude from the corresponding thermodynamic data. Use the Born-Haber cycle and the Kapustinskii equation to calculate the lattice energy.

Useful data:

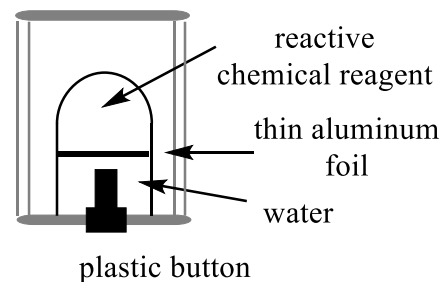
In Cuba, the electricity grid supplies direct current with a voltage of 110 V. Natural gas is composed mostly of methane. The average heat of combustion of wood is 18 kJ/g.

$\Delta H_f^\circ(X)/\text{kJmol}^{-1}$: $\text{H}_2\text{O, liquid} = -286$; $\text{H}_2\text{O, vapor} = -242$; $\text{CH}_4 = -79$; $\text{CO}_2 = -394$;

The charge of an electron is $e = 1,602 \cdot 10^{-19} \text{ C}$; one Faradio is 1 mol of electrons.

The fundamental equation of calorimetry is $Q = m \cdot c \cdot \Delta T$, where c is the heat capacity of the material.

The following values are expressed in [kJ/mol]: $I_1(\text{Ca}) = 590$, $I_2(\text{Ca}) = 1145$, $\Delta H_{\text{EA}}(\text{OH}) = -177$ (electron affinity of the hydroxyl radical), $E(\text{O}=\text{O}) = 495$, $E(\text{H}-\text{H}) = 434$, $E(\text{O}-\text{H}) = 465$, $\Delta H_{\text{sub}}(\text{Ca}) = 178$.



Kapustinskii's equation, U is the lattice enthalpy in kJ/mol, r and z are the radii in pm and the charges of the ions, respectively, $\sum v$ is the total number of ions in the empirical formula, $\sum v = 3$ for $\text{Ca}(\text{OH})_2$.

$$U \approx -1,20 \cdot 10^5 \cdot \frac{\sum v \cdot |z^+| |z^-|}{r^+ + r^-} \cdot \left(1 - \frac{34,5}{r^+ + r^-}\right)$$

The thermochemical radii of calcium and hydroxide ions are 114 and 133 pm, respectively.

Problem 5. Iodine monochloride (15% of total; only 11th and 12th grades)

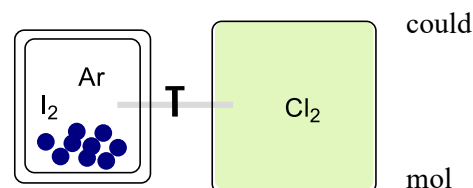
5.1	5.2	5.3	5.4	5.5	5.6	5.7	5.8	5.9	5.10	Total	Total
3	2	4	6	8	22	4	2	7	2	60	15%

Iodine monochloride (ICl) was discovered by the French chemist and physicist Gay-Lussac in 1814. At room temperature, it is a reddish-brown solid that melts and boils at relatively low temperatures. It can be obtained by passing chlorine gas through solid crystals of diiodine.

5.1. In the Faculty of Chemistry laboratory, Juan Carlos (*Yepe*) obtained dichloromethane *in situ* by oxidizing hydrochloric acid with potassium permanganate. What is the equation for the reaction that occurred?

5.2. What essential safety measures should be taken when handling chlorine?

Juan Carlos likes engineering, so he decided to design an industrial process that produce ICl in large quantities. To do so, he designed a reactor A (50 L capacity) connected to a reactor B (100 L capacity), both equipped with a device to control temperature and pressure. A private company decided to implement this design in one of its chemical reagent factories. One day, 50,0 of solid diiodine were introduced into reactor A under an argon atmosphere, at atmospheric pressure and room temperature. The stoichiometric amount of chlorine required for the reaction was introduced into reactor B, also at room temperature. After opening the stopcock connecting the two reactors, the reaction was allowed to continue for a short time. Finally, the product was recovered from the mixture by distillation, and a yield of 86% was calculated.



5.3. Show that the above reaction occurs to its full extent at room temperature by calculating the equilibrium thermodynamic constant.

5.4. Calculate the chlorine pressure inside reactor B before the reaction occurred and the mass of ICl recovered at the end of the process. Assume ideal gas behavior.

5.5. Estimate the temperature at which the purification process by distillation is carried out at atmospheric pressure, that is, the standard boiling point of iodine monochloride.

In an attempt to optimize the process, in another experiment, 1.5 times the amount of chlorine required for the reaction was introduced into reactor B, and the reaction was allowed to continue for longer, until equilibrium was established. The operators were surprised when they detected a lower-than-expected final pressure and bright yellow portions in the solid. Furthermore, the final process yield turned out to be lower. They then contacted Juan Carlos for an explanation. After a literature review, the student told them that iodine trichloride had formed: $\text{Cl}_2(\text{g}) + \text{ICl}(\text{s}) \rightarrow \text{ICl}_3(\text{s})$.

5.6. If all reactions proceeded without any kinetic hindrance, calculate in detail the final pressure within the system (reactors A and B connected) and the amount of each product formed.

5.7. In the solid state, iodine trichloride exists as a dimer. What is the hybridization and coordination number (number of nearest neighbor atoms) of iodine?

5.8. Juan Carlos knows that ICl is used as an iodinating agent in the synthesis of organic compounds because it has a very electrophilic iodine atom with a high partial positive charge. Why is ICl more reactive than diiodine?

One day, Marcos decided to get highlights in his hair, and to do so, he asked his mentor for a bottle of diluted aqueous hydrogen peroxide that was in the lab. But Julito reminded him that he had to determine the concentration before using the solution and suggested using a redox titration. Marcos, being very disciplined, designed the following procedure. First, he transferred a 5,00 mL aliquot of the original peroxide into a 100 mL volumetric flask. After adding distilled water and topping it up to the mark, he took a 20,0 mL aliquot of the new solution and titrated it with a 0,0200 mol/L KMnO_4 standard in the presence of sulfuric acid. The values of consumed titrant were 20,0, 19,8, 19,0 and 19,7 mL.

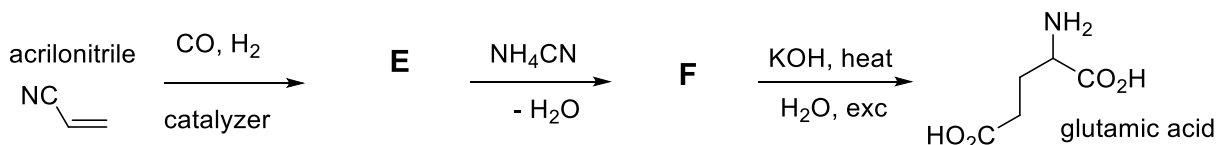
6.5. What is the concentration of the peroxide that Marco found in the laboratory?

6.6. The optimal peroxide content for treating hair is 5,0% (by mass), with a density of 1,02 g/mL. In contrast, the commercial product has a content of 30% and a density of 1,11 g/mL. What volume of commercial peroxide is needed to prepare one liter of the diluted solution?

6.7. After applying the peroxide, Marcos was not satisfied with the result and mistakenly decided to use a more powerful oxidizing agent, hypochlorite, to bleach his hair. Under these conditions, thiols can be oxidized to sulfenic (RSOH), sulfinic (RSO_2H), and sulfonic (RSO_3H) acids. What is the oxidation number of sulfur in these substances? Arrange them in order of increasing acidity and explain your answer.

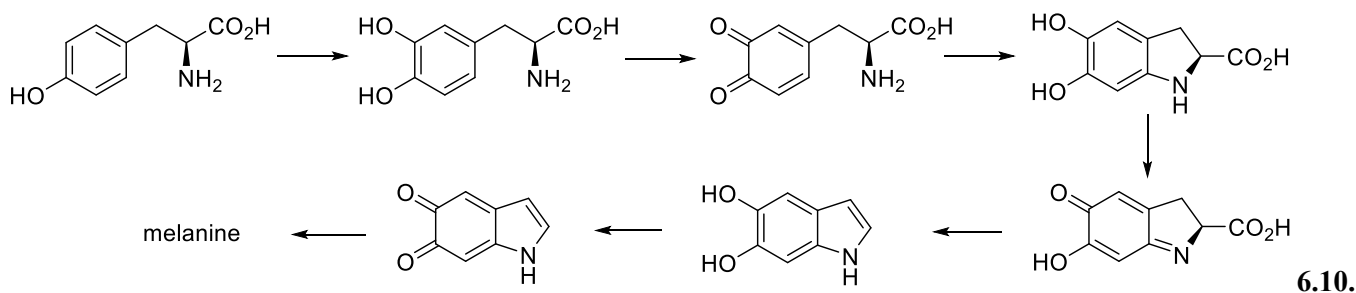
6.8. Hydrogen peroxide is unstable and should be stored at low temperatures and away from light. Formulate the equation for the decomposition reaction of this substance.

Marcos is very boastful and likes to wash his hair with *shampoo* containing keratin. This protein is rich in sulfur and also in the amino acid glutamic acid, which can be obtained from acrylonitrile.



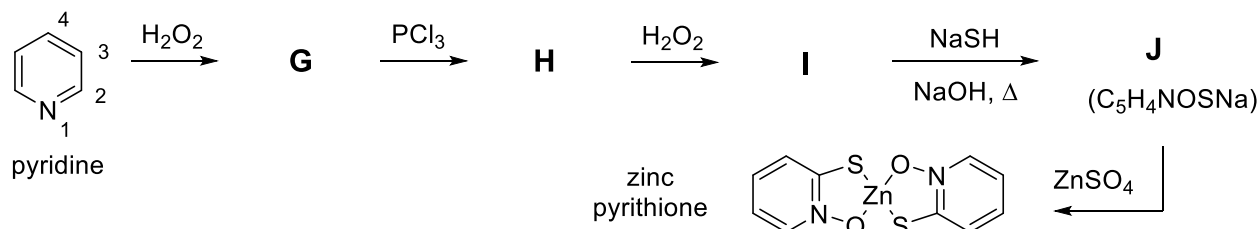
6.9. Represent the structures of synthetic intermediates **E** and **F**.

Hair color is determined by the amount of the pigment melanin. Roy knows that melanins are formed by enzyme-catalyzed reactions from the amino acid tyrosine, according to



Answer questions about the above synthetic sequence: i) How many oxidation steps occur? ii) What is the name of the ring-closing reaction? iii) Why is the indole ring formed in the process aromatic?

To eliminate dandruff, a very annoying fungus that affects the scalp, Roy uses a shampoo whose active ingredient is zinc pyrithione, a synthetic derivative of pyridine.



6.11. Represent the structures of the compounds **G-J**.

Pyridine is an aromatic heterocycle that undergoes S_EAr with strong electrophiles at the 3-position of the ring. In contrast, the corresponding N-oxide **G** favors S_EAr at the 2-position. Halopyridines participate in S_NAr -type reactions more readily when the halogen atoms are located at the 2- and 4-positions (e.g., chloropyridine **H**).

6.12. Represent a relevant (especially stable) structure of the sigma complexes that explain the different regioselectivity in each of the four reactions of S_EAr and S_NAr described above.

ANSWER FROM HERE ON

(Except question 1.1)
